

Project Design Phase
Problem – Solution Fit Template

Date	14 February 2026
Team ID	LTVIP2026TMIDS81120
Project Name	Smart Sorting: Transfer Learning for Identifying Rotten Fruits and Vegetables
Maximum Marks	2 Marks

Problem

Fruit freshness assessment is challenging because it depends on subtle visual features such as color variation, texture changes, and surface patterns that may not always be obvious to the human eye. Traditional inspection methods rely heavily on manual judgment, which can be inconsistent, time-consuming, and prone to error. Environmental factors like lighting conditions and fruit ripeness further complicate accurate evaluation. As a result, it becomes difficult to maintain consistent quality control, leading to food waste, economic loss, and potential health risks. Therefore, there is a need for an intelligent, image-driven system that can automatically analyze fruit characteristics and provide reliable freshness predictions.

Solution

To address this problem, a deep learning–based image classification model is developed to analyze fruit images and predict whether they are fresh or spoiled. Multiple model configurations and training strategies were tested, including variations in convolutional neural network architectures and data augmentation techniques. The models were evaluated using performance metrics such as accuracy and validation loss. After comparison, the best-performing model was selected and integrated into a Flask-based web application that allows users to upload images and receive instant freshness predictions.

Purpose

A convolutional neural network was selected as the final model because of its strong performance in image recognition tasks. CNNs automatically learn hierarchical visual features such as edges, textures, and patterns, enabling accurate classification even with complex image variations. The model demonstrates good generalization ability, minimizes overfitting through augmentation and training controls, and efficiently processes new images in real time. Due to its reliability, scalability, and superior capability in visual analysis, this deep learning approach was chosen as the core engine for the fruit freshness classification system.